

# Galactic Structure, Galaxies & the Interstellar Medium

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|---------------------------------------------------------------------|----------|---------------------------------------------------------------------|-----------|
| <b>Contents</b>                                                     |          | <b>ics</b>                                                          | <b>5</b>  |
| <b>1 Introduction</b>                                               | <b>1</b> | 4.1 Accretion Disk Conversion Efficiency . . . . .                  | 5         |
| <b>2 Deconstructing Binary Systems</b>                              | <b>1</b> | 4.2 The Eddington Luminosity Limit                                  | 5         |
| 2.1 Eclipsing Binaries: Radii from Contact Times . . . . .          | 1        | <b>5 Galactic Architecture and Kinematics</b>                       | <b>6</b>  |
| 2.2 Spectroscopic Binaries: Radial Velocities and the Mass Function | 2        | 5.1 Morphology of the Milky Way .                                   | 6         |
| <b>3 The Interstellar Medium and Interstellar Dust</b>              | <b>2</b> | 5.2 Differential Rotation and the Oort Constants . . . . .          | 6         |
| 3.1 Extinction and Scattering . . . .                               | 2        | 5.3 Galactic Rotation Curves and the Missing Mass Problem . . . . . | 7         |
| 3.2 Color Excess and Intrinsic Colors                               | 3        | <b>6 Galaxies at Large Scales</b>                                   | <b>8</b>  |
| 3.3 Gas-Phase Regions: HI Clouds vs. HII Nurseries . . . . .        | 3        | 6.1 The Hubble Classification System                                | 8         |
| 3.4 The 21 cm Spin-Flip Line . . . .                                | 3        | 6.2 Tully-Fisher and Faber-Jackson Relations . . . . .              | 9         |
| 3.5 Pulsar Dispersion Measure . . . .                               | 4        | 6.3 Active Galactic Nuclei . . . . .                                | 9         |
| <b>4 Accretion Mechanics and Energet-</b>                           | <b>7</b> | <b>7 Summary Sheet</b>                                              | <b>10</b> |

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## Prerequisite Knowledge

- Doppler shift kinematics
- Barycentric coordinate balancing — the center-of-mass condition  $M_1 r_1 = M_2 r_2$
- Right ascension coordinate maps and the general vocabulary of spherical coordinates
- Comfort handling vector transformations (projecting a 3D velocity onto a line of sight)
- Braincells (imported from Andromeda plus tax)

# 1 Introduction

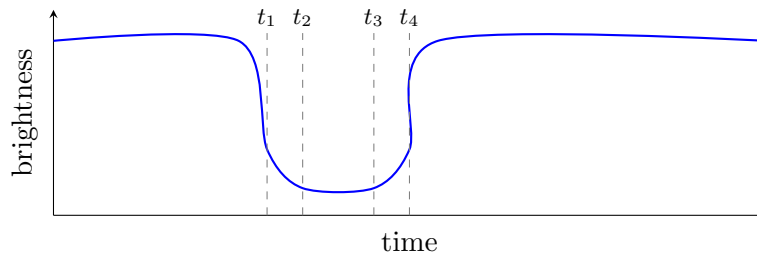
Imagine you're a photon. You were born in the core of a star, in a binary system, and it already took you a hundred thousand years just to fight your way out of your own star's interior. You finally escape into space, feeling pretty good about yourself — and IMMEDIATELY the universe starts toying with you.

First, your own star's companion gets in the way, blocking part of your view and telling anyone watching your light dim on a schedule exactly how big both of you are. Then, on your way out of the system, some of your siblings get Doppler-shifted back and forth so hard that astronomers can weigh your entire star system without ever seeing it directly. Once you're finally out in open space, you don't get to travel in peace either — you go crashing through cloud after cloud of interstellar dust, getting scattered, reddened, and dimmed the whole way, while your radio-wave cousins get delayed differently at every frequency by all the loose electrons drifting between the stars. And this is all *before* you even notice that the entire galaxy around you is rotating, dragging you along a curve that, frankly, shouldn't be possible unless there's a huge amount of invisible mass out there (dark matter, yes.)

## 2 Deconstructing Binary Systems

### 2.1 Eclipsing Binaries: Radii from Contact Times

When a binary's orbital plane happens to lie close to our line of sight, the two stars periodically pass in front of one another, producing dips in the combined light curve. The exact shape of those dips — specifically, the four moments (**contact times**) at which the eclipse begins and ends — turns out to be enough to solve for both stars' actual physical radii.



*Figure 1: An eclipsing binary light curve.  $t_1$ : first contact (limbs just touch).  $t_2$ : second contact (smaller star fully superimposed).  $t_3$ : third contact (egress begins).  $t_4$ : fourth contact (limbs separate completely).*

Set up a coordinate  $x(t)$  for the projected center-to-center separation of the two stars (with  $x = 0$  at mid-eclipse), moving at the constant relative transverse speed  $v = 2\pi a/P$  (valid for a circular, edge-on orbit). Contact happens whenever the two stellar limbs just touch, i.e. whenever  $|x|$  equals the sum or difference of the two radii  $R_l$  (larger star) and  $R_s$  (smaller star):

$$x(t_1) = R_l + R_s, \quad x(t_2) = R_l - R_s, \quad x(t_3) = -(R_l - R_s), \quad x(t_4) = -(R_l + R_s).$$

Since  $x$  changes at the constant rate  $v$ , differences in contact time convert directly into differences in  $x$ :

$$v(t_4 - t_1) = 2(R_l + R_s), \quad v(t_3 - t_2) = 2(R_l - R_s).$$

Adding and subtracting these two equations isolates each radius individually:

$$\boxed{R_l = \frac{v}{4} \left[ (t_4 - t_1) + (t_3 - t_2) \right]}, \quad \boxed{R_s = \frac{v}{4} \left[ (t_4 - t_1) - (t_3 - t_2) \right]}.$$

The relative speed  $v$  itself is fixed once the orbit's semi-major axis  $a$  and period  $P$  are known — and as we're about to see, that's exactly what a spectroscopic orbit solution provides.

## 2.2 Spectroscopic Binaries: Radial Velocities and the Mass Function

For a binary too close to resolve visually, the only handle we have is the Doppler-shifted radial velocity of (at least) one star, oscillating sinusoidally with the orbital period as the star swings toward and away from us. For a circular orbit inclined at angle  $i$  (with  $i = 90^\circ$  meaning perfectly edge-on), the observed radial velocity amplitude of star 1 is

$$K_1 = \frac{2\pi a_1}{P} \sin i,$$

where  $a_1$  is star 1's own orbital radius about the system's center of mass. The center-of-mass condition (Handout 2) requires  $M_1 a_1 = M_2 a_2$ , so  $a_1 = a M_2 / (M_1 + M_2)$ , where  $a = a_1 + a_2$  is the full separation. Substituting this into  $K_1$ , then eliminating  $a$  using Kepler's third law ( $a^3 = G(M_1 + M_2)P^2/4\pi^2$ ), and solving for the combination of masses and inclination that remains, gives the **mass function**:

$$\boxed{f(M) \equiv \frac{M_2^3 \sin^3 i}{(M_1 + M_2)^2} = \frac{PK_1^3}{2\pi G}}.$$

Everything on the right-hand side is directly observable from a single star's spectrum alone (its period and RV amplitude) — but the left-hand side tangles together  $M_1$ ,  $M_2$ , and  $i$  into one combination that can't be separated without more information. Since  $\sin i \leq 1$  always,  $f(M)$  on its own only ever supplies a firm *lower limit* on the companion's mass  $M_2$ .

If *both* stars' spectral lines are visible (a **double-lined** spectroscopic binary), the center-of-mass condition directly gives the mass ratio, since  $M_1 v_1 = M_2 v_2$  implies

$$\boxed{\frac{M_1}{M_2} = \frac{K_2}{K_1}}.$$

Combined with each star's own mass function, this pins down  $M_1 \sin^3 i$  and  $M_2 \sin^3 i$  — *still* degenerate with the unknown inclination  $i$ . This is exactly why the two techniques of this section are so often used together: a system that is *both* eclipsing and double-lined spectroscopic hands us  $i \approx 90^\circ$  directly from the eclipse geometry, breaking the degeneracy and yielding true, absolute masses *and* radii for both stars — one of the only truly direct, model-independent ways to weigh and measure a star.

## 3 The Interstellar Medium and Interstellar Dust

### 3.1 Extinction and Scattering

Starlight traveling through the Galaxy is dimmed and reddened by microscopic dust grains, via a combination of absorption and scattering collectively called **extinction**. Because these grains are comparable in size to (or smaller than) the wavelength of visible light, they scatter shorter (bluer)

wavelengths considerably more efficiently than longer (redder) ones. Empirically, over the optical range, the extinction in magnitudes per unit path length scales roughly as

$$A_\lambda \propto \frac{1}{\lambda}.$$

The practical consequence is **reddening**: a star seen through a dusty line of sight looks both fainter *and* redder than it intrinsically is, since more of its blue light has been scattered out of the line of sight than its red light.

### 3.2 Color Excess and Intrinsic Colors

Define the **color excess** as the extra reddening in a particular pair of bands, e.g.

$$E(B - V) \equiv (B - V)_{obs} - (B - V)_0,$$

the difference between a star’s observed color and its true, dust-free (intrinsic) color. Since extinction affects both the total brightness and the color simultaneously, the two are linked through the **total-to-selective extinction ratio**  $R_V \approx 3.1$  (a typical Milky Way average):

$$A_V = R_V E(B - V).$$

Once  $E(B - V)$  is known (usually estimated by comparing a star’s observed color to the intrinsic color expected for its spectral type), both the star’s true brightness and its true color can be recovered:

$$m_{V,0} = m_V - A_V, \quad (B - V)_0 = (B - V)_{obs} - E(B - V).$$

Skipping this correction is a regular source of systematic error — an obscured star can easily be mistaken for one that’s simply farther away, cooler, or both.

### 3.3 Gas-Phase Regions: H I Clouds vs. H II Nurseries

The gas between the stars is not uniform — it exists in starkly different phases depending on its local radiation environment:

- **H I regions**: cold ( $\sim 100\text{ K}$ ), neutral atomic hydrogen, far from any hot ionizing star. This is the ordinary, quiescent bulk of the Galaxy’s interstellar gas.
- **H II regions**: hot ( $\sim 10^4\text{ K}$ ), ionized hydrogen surrounding young, massive O and B stars, whose intense ultraviolet output (photon energy exceeding hydrogen’s  $13.6\text{ eV}$  ionization threshold) strips electrons from every atom within a characteristic volume (a **Strömgren sphere**). Because H II regions only exist around young, massive, short-lived stars, they double as signposts of *active, ongoing* star formation.

### 3.4 The 21 cm Spin-Flip Line

Ordinary neutral hydrogen (H I) is essentially invisible in optical light — its electronic transitions all require far more energy than the cold gas can supply. But the hydrogen atom’s ground state has a subtler internal structure: the proton and electron each carry spin, and the two configurations — spins aligned (slightly higher energy) versus anti-aligned (slightly lower energy) — differ by a tiny amount of **hyperfine** energy. A spontaneous “flip” from the aligned to anti-aligned state releases a photon at the corresponding wavelength of  $21\text{ cm}$  (in the radio band).

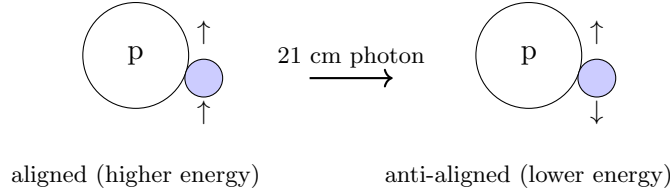


Figure 2: The 21 cm hyperfine spin-flip transition: the electron's spin flips relative to the proton's, releasing a low-energy radio photon.

This transition is enormously *forbidden* — a given atom waits, on average, some  $\sim 10^7$  years before spontaneously flipping — but the sheer quantity of neutral hydrogen spread across the Galaxy makes the line easily detectable anyway. Because 21 cm radiation passes through dust completely unimpeded (unlike optical light, per Section 2.1), and because its observed Doppler shift directly encodes the gas's line-of-sight velocity, this single spectral line is our single most important tool for mapping the distribution *and* rotation of gas throughout the Galaxy, including regions totally hidden from optical view behind the Galactic disk's own dust.

### 3.5 Pulsar Dispersion Measure

A radio pulse traveling through the ionized interstellar medium doesn't travel at a fixed speed independent of frequency, the way light does in a vacuum — free electrons in the plasma make the medium dispersive. The group velocity of a radio wave of frequency  $\nu$  passing through a plasma of electron density  $n_e$  is

$$v_g \approx c \left( 1 - \frac{\nu_p^2}{2\nu^2} \right), \quad \nu_p^2 \propto n_e,$$

where  $\nu_p$  is the local plasma frequency. Lower frequencies are slowed down more than higher ones. Integrating the resulting extra travel time along the entire path length  $L$  to the pulsar, and defining the **dispersion measure**  $DM \equiv \int_0^L n_e dl$ , the arrival-time delay between two observing frequencies  $\nu_1 < \nu_2$  works out to

$$\Delta t = \frac{e^2}{2\pi m_e c} DM \left( \frac{1}{\nu_1^2} - \frac{1}{\nu_2^2} \right).$$

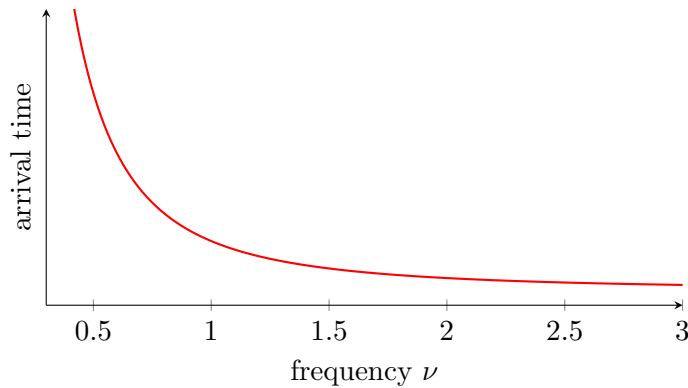


Figure 3: A pulsar's characteristic dispersion “sweep”: lower frequencies arrive measurably later than higher ones, delayed by an amount  $\propto 1/\nu^2$ .

Since  $\Delta t$  is directly measurable (by comparing pulse arrival times across a radio receiver's bandwidth) and every constant in the formula is known, this equation is solved directly for  $DM$ . Given

an independent model of the Galaxy’s free-electron density distribution  $n_e(\vec{r})$  (built up from many pulsars with otherwise-known distances), a newly discovered pulsar’s own  $DM$  can then be inverted into a distance estimate — one of the very few methods available for gauging distances to objects buried deep in the Galactic plane.

## 4 Accretion Mechanics and Energetics

### 4.1 Accretion Disk Conversion Efficiency

When matter spirals inward onto a compact object (a white dwarf, neutron star, or black hole) from far away, it releases gravitational potential energy as it falls. Dropping a mass  $m$  from effectively infinite distance down to the object’s surface (or innermost stable orbit) at radius  $R$  releases

$$\Delta E \sim \frac{GMm}{R}.$$

The efficiency of this process — energy released per unit rest-mass energy — is therefore

$$\boxed{\eta \sim \frac{GM}{Rc^2}.$$

For a neutron star or black hole,  $R$  is only a handful of Schwarzschild radii (Handout 5, Section 5.4), making  $\eta$  of order 0.1–0.4 — accretion onto a compact object converts 10–40% of the infalling rest mass directly into radiated energy, compared to nuclear fusion’s mere  $\sim 0.7\%$  (the fraction of mass converted to energy in hydrogen-to-helium fusion). This is exactly why accreting compact objects — X-ray binaries, and ultimately active galactic nuclei (Section 5.3) — are among the most efficient, and therefore most luminous, power sources in the universe.

### 4.2 The Eddington Luminosity Limit

There is a ceiling on how luminous an accreting object can be while still able to hold onto its own infalling matter: at high enough luminosity, the outward push of radiation pressure on the infalling gas (via electron scattering) simply overwhelms gravity’s inward pull, blowing the material away before it can accrete at all.

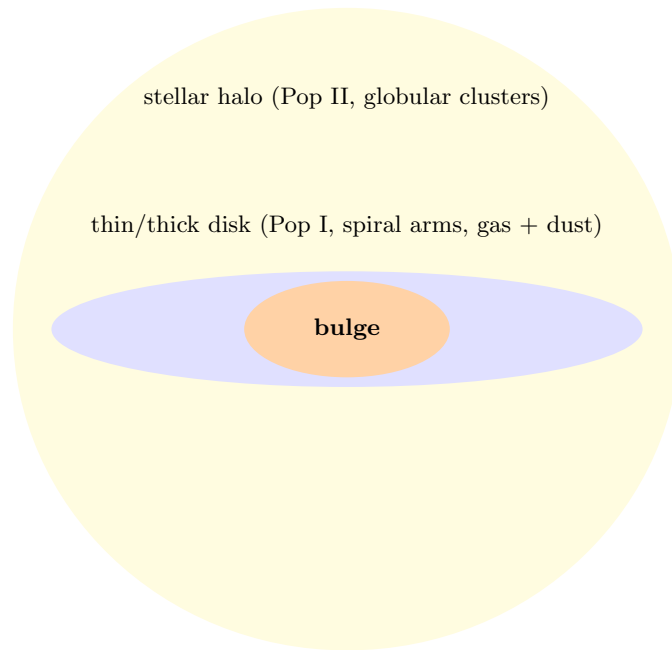
The outward radiation force per unit mass of gas, from a source of luminosity  $L$  at distance  $r$ , using opacity  $\kappa$  (the effective cross-section for radiation per unit mass) is  $\kappa F/c = \kappa L/(4\pi r^2 c)$ . The inward gravitational force per unit mass is simply  $GM/r^2$ . Setting these equal defines the critical (**Eddington**) luminosity, above which radiation wins and accretion is choked off:

$$\frac{\kappa L_{Edd}}{4\pi r^2 c} = \frac{GM}{r^2} \implies \boxed{L_{Edd} = \frac{4\pi GMc}{\kappa}.$$

Notice that the distance  $r$  cancels completely — exactly as it did for the radiation-pressure balance in Handout 4 (Section 2.5), since both forces dilute identically with distance. The Eddington limit therefore depends only on the accreting object’s own mass (and the gas’s opacity), making it an extremely useful benchmark: an observed luminosity anywhere near  $L_{Edd}$  for a given mass is a strong hint that the object is accreting close to its own physical limit.

## 5 Galactic Architecture and Kinematics

### 5.1 Morphology of the Milky Way



*Figure 4: Schematic cross-section of the Milky Way's major components.*

- **Disk:** a flattened structure containing the spiral arms, most of the Galaxy's gas and dust, and young, metal-rich **Population I** stars on roughly circular orbits.
- **Bulge:** a dense, roughly spheroidal (sometimes boxy or peanut-shaped) concentration of older stars near the Galactic center, often hosting a central bar.
- **Halo:** a sparse, roughly spherical distribution of very old, metal-poor **Population II** stars and globular clusters, on highly eccentric, randomly oriented orbits — along with the much larger, invisible dark matter halo that dominates the Galaxy's total mass (Section 4.3).

The Population I/II distinction is more than just age: Population I stars formed from gas already enriched by earlier generations of stars, so they carry higher heavy-element (“metal”) abundances, while Population II stars, formed early in the Galaxy's history before much enrichment had occurred, are correspondingly metal-poor.

### 5.2 Differential Rotation and the Oort Constants

Unlike a rigid body, the Galaxy rotates **differentially** — different radii orbit at different angular speeds  $\Omega(R) = \Theta(R)/R$ , where  $\Theta(R)$  is the circular orbital speed at radius  $R$ . This has directly observable local consequences for how nearby stars appear to move relative to the Sun.

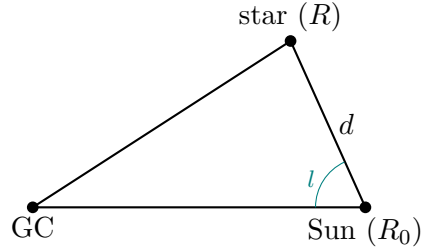


Figure 5: The Sun-star-Galactic Center triangle used to derive the Oort constants.  $R_0$  is the Sun's galactocentric distance,  $d$  the Sun-star distance, and  $l$  the Galactic longitude of the star.

*Sketch of derivation:* Taylor-expand the angular speed of a nearby star about the Sun's own value,  $\Omega(R) \approx \Omega_0 + \left(\frac{d\Omega}{dR}\right)_0 (R - R_0)$ , and project the resulting relative velocity between Sun and star onto the line of sight (giving the observed radial velocity) and perpendicular to it (giving the observed transverse/proper motion), using the triangle in Figure 5 to relate  $R$ ,  $d$ , and  $l$ . Keeping only first order in  $d/R_0$ , the results take the elegant form

$$v_r = Ad \sin 2l, \quad v_t = Ad \cos 2l + Bd,$$

where  $A$  and  $B$  are the **Oort constants**:

$$A = \frac{1}{2} \left( \frac{\Theta_0}{R_0} - \left( \frac{d\Theta}{dR} \right)_0 \right), \quad B = -\frac{1}{2} \left( \frac{\Theta_0}{R_0} + \left( \frac{d\Theta}{dR} \right)_0 \right).$$

Two combinations of these constants carry direct physical meaning:  $A - B = \Theta_0/R_0 = \Omega_0$ , the Sun's own local angular orbital speed, and  $A + B = -\left(\frac{d\Theta}{dR}\right)_0$ , the local shear (how sharply the rotation curve's speed changes with radius). Measured values in the solar neighborhood,  $A \approx 15 \text{ km/s/kpc}$  and  $B \approx -13 \text{ km/s/kpc}$ , immediately confirm that the Galaxy is not rotating rigidly ( $A \neq 0$ ).

### 5.3 Galactic Rotation Curves and the Missing Mass Problem

If essentially all of the Galaxy's mass were concentrated within its visible disk, we'd expect orbital speeds far outside that disk to fall off in a simple Keplerian way,  $\Theta(r) \propto r^{-1/2}$  (exactly as planetary orbital speed falls with distance from the Sun). Instead, observed rotation curves — traced out to great distances using the 21 cm line (Section 2.4) — stay remarkably **flat**,  $\Theta(r) \approx \text{const}$ , far beyond the edge of the visible stars and gas.

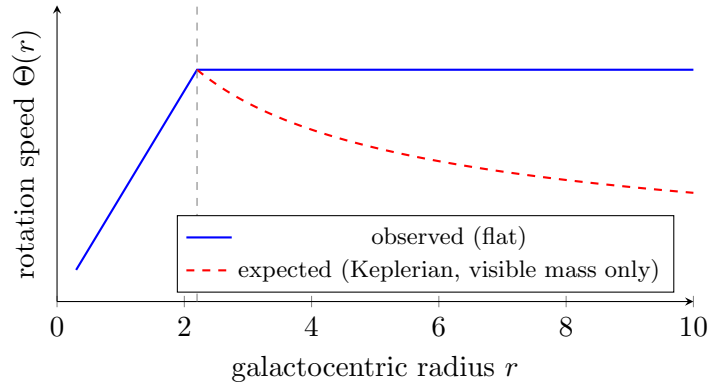


Figure 6: Observed (flat, blue) vs. expected Keplerian (falling, red dashed) rotation curves. The discrepancy beyond the visible edge is the missing mass problem.

A flat rotation curve is a direct statement about the mass distribution. Circular orbital balance requires  $\Theta^2/r = GM(r)/r^2$ , so

$$M(r) = \frac{\Theta^2 r}{G}.$$

If  $\Theta$  is constant,  $M(r) \propto r$  — the enclosed mass keeps growing linearly with radius, far past the point where any visible starlight remains. Differentiating to get the corresponding density profile, using mass conservation  $dM/dr = 4\pi r^2 \rho(r)$  from Handout 5,

$$\frac{dM}{dr} = \frac{\Theta^2}{G} = 4\pi r^2 \rho(r) \implies \boxed{\rho(r) \propto \frac{1}{r^2}}.$$

This is the density profile of a (idealized) **singular isothermal sphere** — an enormous, invisible, roughly spherical halo of matter whose density falls off just slowly enough to keep piling up more and more mass at ever-larger radii. Since this extra mass emits no detectable light at any wavelength, it's known as **dark matter**, and rotation curves like this one remain some of the most direct observational evidence for its existence.

## 6 Galaxies at Large Scales

### 6.1 The Hubble Classification System

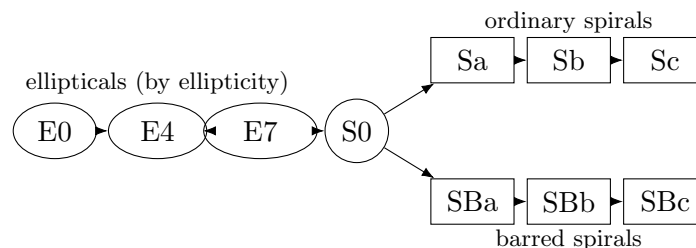


Figure 7: The Hubble “tuning fork” diagram. Despite its historical framing as an evolutionary sequence, it is now understood purely as a morphological classification — galaxies don’t literally evolve from left to right. (would have been cool though.)

- **Ellipticals** (E0–E7): smooth, featureless light profiles, classified purely by apparent ellipticity,  $\epsilon = 1 - b/a$  (E0 nearly round, E7 highly flattened); little gas or dust, mostly old stars, supported by random stellar orbital motions (velocity *dispersion*) rather than ordered rotation.
- **Spirals** (Sa–Sc, or barred SBa–SBc): a flattened disk with spiral arms wound progressively looser from Sa (tightly wound, large bulge) to Sc (loosely wound, small bulge); rich in gas, dust, and ongoing star formation; supported primarily by ordered, near-circular rotation.
- **Lenticulars** (S0): an intermediate type with a disk but no spiral arms and very little gas left to form new stars.
- **Irregulars**: no organized symmetric structure at all, often gas-rich and rich in active star formation, frequently a consequence of tidal disturbance or an ongoing merger.

## 6.2 Tully-Fisher and Faber-Jackson Relations

Both spirals and ellipticals obey a strikingly tight empirical relationship between their total luminosity and a single kinematic measurement — immensely useful, since it turns an easily measured velocity into a stand-in for distance (via the inverse-square law, once the true luminosity is inferred).

For a self-gravitating stellar system in virial equilibrium (Handout 5, Section 4.2),  $GM/R \sim v^2$ , where  $v$  is the characteristic internal velocity (rotation speed for a spiral, velocity dispersion for an elliptical). If, further, galaxies of a given type all share roughly the same mass-to-light ratio ( $M \propto L$ ) and roughly the same characteristic surface brightness ( $\Sigma \equiv L/R^2 \approx \text{const}$ , so  $R \propto L^{1/2}$ ), then substituting both scalings into the virial relation,

$$v^2 \sim \frac{GM}{R} \propto \frac{L}{L^{1/2}} = L^{1/2} \implies \boxed{L \propto v^4}.$$

Applied to a spiral galaxy's rotation speed  $v_{rot}$  (typically measured from the width of its 21 cm line profile), this is the **Tully-Fisher relation**; applied to an elliptical galaxy's stellar velocity dispersion  $\sigma$ , it is the **Faber-Jackson relation**. Both are, at heart, the exact same virial-plus-constant-surface-brightness argument, just applied to a rotation-supported system in one case and a dispersion-supported one in the other.

## 6.3 Active Galactic Nuclei

Some galaxies have a central point-like source vastly outshining the rest of the entire galaxy combined — an **active galactic nucleus (AGN)**. The energy source in every case is the same: gas accreting (Section 3) onto a **supermassive black hole** ( $10^6$ – $10^{10} M_\odot$ ) at the galactic center, radiating close to (or occasionally even briefly above) its own Eddington limit.

The bewildering variety of observed AGN types turns out to be largely a matter of **viewing geometry** and luminosity, not fundamentally different physical objects:

- **Seyfert galaxies**: relatively low-luminosity AGN in otherwise ordinary, nearby spiral host galaxies, bright enough to be noticed against their host but not to outshine it entirely; subdivided into Type 1 (broad *and* narrow emission lines, when we have a relatively unobstructed view of the fast-moving gas close to the black hole) and Type 2 (narrow lines only, when a surrounding dusty torus blocks our direct view of the innermost, broad-line region).
- **Quasars**: the extreme, high-luminosity end of the same phenomenon, luminous enough to completely outshine their host galaxy and be seen across cosmological distances.
- **Blazars**: AGN whose relativistic jet happens to point almost directly at Earth; the resulting relativistic beaming dramatically boosts the observed brightness and produces the extreme, rapid variability characteristic of this subclass.

In short: same central engine (an accreting supermassive black hole near its Eddington limit), different combinations of luminosity, host galaxy prominence, and the lucky (or unlucky) angle from which we happen to be looking down the barrel of the jet.

## 7 Summary Sheet

| Concept                                | Formula                                                                                                                                       |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Eclipsing binary radii                 | $R_l = \frac{v}{4}[(t_4 - t_1) + (t_3 - t_2)], R_s = \frac{v}{4}[(t_4 - t_1) - (t_3 - t_2)]$                                                  |
| Spectroscopic mass function            | $f(M) = \frac{M_2^3 \sin^3 i}{(M_1 + M_2)^2} = \frac{PK_1^3}{2\pi G}$                                                                         |
| Double-lined mass ratio                | $M_1/M_2 = K_2/K_1$                                                                                                                           |
| Extinction law (empirical)             | $A_\lambda \propto 1/\lambda$                                                                                                                 |
| Color excess / total extinction        | $A_V = R_V E(B - V), R_V \approx 3.1$                                                                                                         |
| Pulsar dispersion delay                | $\Delta t = \frac{e^2}{2\pi m_e c} DM \left( \frac{1}{\nu_1^2} - \frac{1}{\nu_2^2} \right)$                                                   |
| Accretion efficiency                   | $\eta \sim GM/(Rc^2)$                                                                                                                         |
| Eddington luminosity                   | $L_{Edd} = 4\pi GMc/\kappa$                                                                                                                   |
| Oort constants                         | $A = \frac{1}{2} \left( \frac{\Theta_0}{R_0} - \dot{\Theta}_0 \right), B = -\frac{1}{2} \left( \frac{\Theta_0}{R_0} + \dot{\Theta}_0 \right)$ |
| Local angular velocity / shear         | $A - B = \Omega_0, A + B = -\dot{\Theta}_0$                                                                                                   |
| Dark matter halo (flat rotation curve) | $M(r) \propto r, \rho(r) \propto 1/r^2$                                                                                                       |
| Tully-Fisher / Faber-Jackson           | $L \propto v_{rot}^4 / L \propto \sigma^4$                                                                                                    |